RUDRA PRASADH GANESH

grudraprasadh@gmail.com · github.com/Unknown-Blaze · linkedin.com/in/unknown-blaze · unknown-blaze.github.io/portfolio

Full Stack Software Engineer building scalable web platforms, low-latency APIs, and reliable cloud-native systems

EDUCATION

Nanyang Technological University, Singapore **Aug 2022 – May 2026**
Bachelor of Engineering (Computer Engineering) CGPA: 4.94 / 5.00

- Graduated with **Honours (Highest Distinction)** — the highest classification NTU awards.
- **Lee Kuan Yew Gold Medal** – awarded to the top graduating student of the cohort.
- NTU Science and Engineering Undergraduate **Scholar** – Full Scholarship.
- Achieved **Dean's List** in all four academic years; **Top 5%** of the cohort every year.

Sri KDU International School, Malaysia **Sep 2020 – May 2022**
IB Diploma – 100% Tuition Scholar Score: 42 / 45

SKILLS

- **Languages:** Python, TypeScript/JavaScript, Java, C#, SQL, Bash, HTML/CSS, C/C++
- **Backend & APIs:** FastAPI, Django/DRF, Node.js/Express, ASP.NET, Spring Boot, REST, WebSockets, OAuth/JWT
- **Frontend:** React, Next.js, React Native/Expo, Redux, Tailwind CSS, Vite/Webpack
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Firebase/Firestore
- **Cloud & DevOps:** Alibaba Cloud, AWS (Lambda, S3, Elastic Beanstalk), Docker, Kubernetes, Terraform, CI/CD
- **AI/ML:** PyTorch, TensorFlow, Hugging Face, Scikit-Learn, RAG (LangChain/LangGraph), Ollama
- **Blockchain & Web3:** Solidity, Hardhat, Ethers.js/Web3.js, IPFS, Ethereum, XRPL
- **Tools:** Git/GitHub, Linux, JIRA, Postman

EXPERIENCE

Fabrica AI **Jul 2026 – Present**
Software Engineer *Singapore*

- **Sole software engineer**, owning the entire software function at a robotics startup after the tech lead's departure — architecture, delivery, and code review across every workstream.
- **Direct intern engineers** and integrate across mechanical and electrical verticals to build a modular **ROS2** codebase for a fully autonomous tile-grouting robot shipping in Singapore and Australia.
- Delivered a full **LiDAR** sensor upgrade from integration to on-site deployment in **one month**.
- Containerize dev and on-robot builds with **Docker**, automate CI/CD with **GitHub Actions**, support hiring.

Lumerva **Sep 2025 – Present**
Co-Founder & Full Stack Engineer *Singapore*

- Built a production learning platform with **React + TypeScript** frontend and Python backend services.
- Designed asynchronous processing with **Celery + Redis** for background jobs, reducing request-path bottlenecks.
- Integrated external content APIs (**YouTube API**, **Brave API**) with ranking logic to improve discovery quality.

Misumi Kaizen OÜ Group (MISUMI.AI) **Sep 2025 – Aug 2026**
DevOps Engineer & Team Lead (veNTUre Project) *Singapore (Hybrid)*

- Led **LISA 1.0**, an automotive advisory platform in production with MISUMI's international customers.
- Architected containerized microservices on **Alibaba Cloud** with a **Terraform** pipeline across **4 ASEAN markets**.
- Cut API latency from **300ms** to **50ms** and cloud spend by **70%** via caching and rightsizing.
- Handed LISA 1.0 to an incoming team and continued on **LISA 2.0** as developer and technical advisor.

MODEC Offshore Production Systems **Dec 2025 – Jan 2026**
DevOps Intern *Singapore*

- Owned deployment of a sustainability data application from PoC to secure production on **AWS Elastic Beanstalk**.

NTU, National Institute of Education **Jun 2025 – Oct 2025**
AI Engineer (Part-Time) *Singapore (Hybrid)*

- Architected a real-time **WebSocket** platform achieving **99.9%** session stability.
- Built internal tooling for session diagnostics and reliability testing.
- Co-authored a **peer-reviewed Springer paper** (AIED 2026) from the resulting **60-participant** study.

MODEC Offshore Production Systems **Jun 2025 – Aug 2025**
Full Stack Software Engineer Intern (Sustainability Analytics) *Singapore (On-site)*

- Built a React/Django dashboard and cut data retrieval time **40%** with **Redis** caching.
- Designed REST APIs handling **500+ concurrent requests** at **99.9%** uptime.
- Automated ingestion via **AWS Lambda + S3** to integrate IoT logs into analytics datasets.

Panasonic R&D Center **Jan 2025 – Jun 2025**
Backend / Full Stack Engineering Intern *Singapore (On-site)*

- Deployed backend services (**FastAPI + Docker**) with async request handling.
- Built a resilient data pipeline with SQL, Python, and Redis, improving throughput and cache hits.
- Implemented monitoring and iteration loops to track failures and improve service quality.

Schaeffler Hub for Advanced Research **May 2024 – Aug 2024**
Embedded Engineering Intern (Edge Data Systems) *Singapore (On-site)*

- Integrated **CAN bus** master–slave control between controller and actuator on an **XMC4800** MCU.
- Optimized performance **20%** via low-latency **EtherCAT** and **FreeRTOS** scheduling.
- Enabled higher-frequency telemetry for predictive maintenance and edge analytics.

HACKATHONS

nGoDonate (EasyA x Ripple Hackathon) **Jun 2025 – Nov 2025**
Winner — Best Developer Tooling · On-chain incentives + donation escrow *Source Code*

- Built a donation workflow on **XRPL** and **Ethereum** with milestone escrow and **2s** finality.

FunctionGemma (Cactus x Google Global Hackathon) **Feb 2025**
National Top 10 — Workflow automation platform prototype *Source Code*

- Engineered event-driven workflow orchestration with modular, reusable **Python API** service boundaries.

- **Also: Top 5** TechFest (deepfake detection); **Finalist** DSTA BrainHack and PSA Code Sprint.

PROJECTS

MHC-Peptide Binding Prediction — Final Year Project **Aug 2025 – May 2026**
Multimodal protein modeling for MHC-peptide affinity prediction *Bioinformatics Lab*

- Built the dataset from **IEDB**, generating 3D structures via a **self-developed algorithm extending ColabFold**, with **homology-based 80/10/10 splits** to prevent sequence-similarity leakage.
- Retrained competing SOTA models on the same data and split, beating the strongest: **PCC 0.859 vs. 0.847**; R^2 and Spearman up 0.5–2%.
- Fused **ESM-2 (650M)** embeddings with the structural pipeline under **LoRA/PEFT** in **PyTorch** on an **A100 (80GB)**.

MyCCDS Student Platform **Nov 2024 – Present**
Production iOS and Android app for NTU CCDS students *App Store* | *Play Store*

- Shipped a **React Native/Expo** app with **600+ daily active users** and a **5-star** store rating.

Mango Order Management System **Jan 2025 – May 2025**
B2B ordering, payments, and operational analytics for a retailer *Live Demo*

- Built end-to-end workflows (**React, Firebase, Stripe**) processing **SGD18,000+ monthly** in transactions.

President Research Scholar — URECA **Aug 2024 – Apr 2025**
Edge AI security pipeline on ODROID *URECA Office*

- Achieved **96%** motion detection accuracy at **180ms** latency on **ODROID**.
- Tuned pipeline thresholds and buffering to cut false positives under changing lighting.

Leadership & Extracurriculars: President/Captain, NTU International Chess Varsity Team • Music Director, Piano Ensemble • NTU Peer Coach • Malaysia National Cubing Open Finalist • ABRSM Piano Grade 8 (Distinction)